

SCOPE

Reviewed RT18 source routes, order and customer APIs, order details, admin order system, PDF utilities, product fields, database access, package configuration, and branding assets. No application files, schema, dependencies, or configuration were modified.

1. EXISTING INVOICE GENERATION

Invoice generation does not currently exist. There is no customer invoice, invoice download, invoice API, invoice numbering, billing record, GST/tax calculation, or order receipt generation.

2. EXISTING PDF FUNCTIONALITY

RT18 has shipping-label PDF generation only. The single-label page is `app/admin/orders/label/[orderId]/page.tsx`. The bulk route is `app/api/admin/shipping-labels/bulk/route.ts`. Shared rendering is in `lib/shipping-label.ts`. Existing PDF output is a shipping label, not an invoice.

The project already includes `jspdf`. No source implementation uses `ReportLab` or `PDFKit`.

3. DATA AVAILABLE FOR AN INVOICE

Order data: `id`, `created_at`, `user_id`, `customer_name`, `phone`, `address`, `city`, `pincode`, `payment_method`, `payment_status`, `transaction_id`, `payment_proof`, `total_amount`, `order_status`, `cancellation_reason`, and `payment_rejection_reason` where applicable.

Order item data: `order_id`, `product_id`, `quantity`, and `price`. Joined order APIs also return `product_name` and `image_url`.

Customer profile data: `id`, `name`, `email`, `phone`, `address`, `city`, and `pincode`. The order customer snapshot is preferable for an invoice because it represents the details used at purchase time.

Product data: `id`, `name`, `price`, `discount_price`, `stock`, `image_url`, `category_id`, `category_name` through the category join, and `created_at` as used by product queries.

No SKU, HSN, GSTIN, tax rate, or tax classification was found.

4. SUBTOTAL, SHIPPING, AND PAYMENT ADJUSTMENT

Only final `total_amount` is stored. Separate subtotal, shipping, discount, and payment-adjustment fields are not stored.

Current calculation in `app/api/orders/route.ts`: selling price equals price minus `discount_price`; subtotal equals selling price multiplied by quantity; shipping is zero when subtotal is greater than 500, otherwise 70; PREPAID adjustment is minus 20; `total_amount` is the calculated final amount.

A basic invoice can reconstruct these values from order items, but the historical breakdown is not independently persisted.

5. GST AND TAX DATA

No GST or tax data exists in the current source or API contracts. Missing information includes seller legal name and address, seller GSTIN, customer GSTIN, HSN or SAC code, tax rate and tax amount, CGST, SGST, IGST, place of supply, tax-inclusive or tax-exclusive pricing rules, and immutable historical tax snapshots.

A legally complete tax invoice cannot be reliably produced from the current data.

6. INVOICE NUMBERING

Invoice numbering does not exist. There is no invoice number field, sequence, prefix, issue date, invoice status, or duplicate-prevention mechanism. The order ID is available as an order reference but should not automatically be treated as a formal invoice number without an explicit business decision.

7. WHAT IS MISSING

For a basic downloadable order invoice: invoice PDF layout and rendering, customer-authenticated invoice download route, ownership verification, filename convention, seller identity and contact details, invoice date, order reference, item lines, quantities, prices, totals, subtotal, shipping, adjustment, final total, payment method, and payment status.

For a proper tax invoice, persistent invoice numbering, tax fields, GST/HSN data, seller/customer billing data, and historical price/tax snapshots are also required.

8. RECOMMENDED FILES

Basic non-tax invoice:

- app/api/auth/my-orders/[orderId]/invoice/route.ts: authenticated customer invoice download endpoint with ownership checks.
- lib/invoice.ts: shared PDF generation logic.
- app/account/orders/[orderId]/page.tsx: optional invoice download control, subject to UI approval.
- Optional admin invoice endpoint protected by getAdmin().

The existing jsPDF dependency may be reusable, subject to server-runtime compatibility testing.

9. DATABASE SCHEMA REQUIREMENT

A basic reconstructed invoice does not strictly require schema changes. It can use existing order, order-item, customer, and product data and recalculate subtotal and shipping.

A proper tax or legally auditable invoice should use schema-backed data for invoice number, issue date, tax values, HSN/SKU, seller/customer billing details, historical subtotal/shipping/discount breakdown, and immutable snapshots. Schema changes would therefore likely be required for that scope.

BRANDING

The existing RT18 logo is app/assets/brand_new.png. It is already used by site pages, the admin shipping-label page, and bulk shipping-label generation. The shipping-label implementation is the closest existing reference for RT18 PDF branding and jsPDF setup.

FINAL ASSESSMENT

RT18 currently supports shipping-label PDFs only. It has enough data for a basic reconstructed order invoice, but it lacks the tax, GST, SKU/HSN, invoice-number, and immutable accounting data needed for a proper tax invoice.

AUDIT STATUS

Complete. No project source, UI, database, package, configuration, commit, or push changes were made.